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Combined Objective Optimization for Vehicle Routing Using Genetic Algorithm

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Abstract

Vehicle Routing Problem (VRP) is an imperative segment in logistics handling. It helps the transportation and distribution companies in routing by aiding to establish a proper balance between the number of vehicles and total distance. In this work combined objective capacitated vehicle routing problem (CVRP) is considered. CVRP is a combinatorial optimization problem in which a fleet of vehicles with limits on capacity are available to service a set of customers from a central depot with their individual demands known in advance. Genetic Algorithm technique is adopted to solve the problem. Performance of GA's largely depends on the type of genetic operators used. In this work two different crossovers are selected and an attempt is made to bring out the impact of these selected crossovers on the quality of solutions generated. Since the focus is on two objectives of VRP i.e. the number of vehicles and the distance, the fitness assignment procedure with GA known as Fitness Aggregated Genetic Algorithm (FAGA) is employed. The model is validated using information obtained from a distribution firm. The result suggests that the proposed algorithm is highly competitive and extremely efficient for combined objective optimization of vehicle routing.

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Keywords: Vehicle routing problem; Genetic algorithm; Combined objective optimization; Best Cost Route Cross Over (BCRC); Swap Mutation; Aggregate Fitness Value.

1. Introduction

The optimal/efficient utilization of transport of materials and finished products is very much required for the efficient management of supply chain. It is found that most of the organizations find it difficult to find a balance between the number of routes and the total distance to be travelled in fulfilling their delivery requirements,

particularly when the customer base is large. So this paper mainly aims at finding a solution to the transportation/routing problem faced by the transportation and distribution companies and other companies of its kind.

Vehicle routing problem (VRP) is an important topic of research in the field of transportation, logistics and distribution. VRP consists of a set of customers with their individual demands known. Each vehicle starts from the central depot, delivers goods to the respective customers and returns to the same depot. The total demand of any route should be well within the vehicle capacity. Each customer must be serviced by exactly one vehicle. The main objective of VRP is to deliver to a set of customers with known demands optimizing the total distance travelled and the number of vehicles [1]. The capacitated vehicle routing problem (CVRP) is a variant of VRP and is not at all easy to solve. In CVRP's all the customers are served using homogeneous vehicles. Since both the number of vehicles as well the distance need to be optimized simultaneously the problem will be dealt as a combined objective model [2].

In this work Fitness Aggregated Genetic Algorithm (FAGA) is implemented to solve the combined objective CVRP. Performance of GA's largely depends on the type of genetic operators used in it. In this work a comparison is made between two different crossover operators. Padmabati Chand and J.R. Mohanty [3] specify that since their initial attempts to solve a multi objective vehicle routing problem with standard crossover operators like PMX crossover, Uniform orders crossover (UOX) etc. yielded non-competitive solutions, they utilized a problem specific crossover operator called as SMCM crossover which fetched them superior results. V. Sivaram Kumar *et.al* [4] in their attempt to solve a multi objective CVRP, point out that standard genetic operators like one point crossover, two point ordered crossover, uniform ordered crossover, partially mapped crossover and rout crossover generates infeasible solutions for CVRP. Due to this reason they implemented another specialized cross over known as Best Cost Route Crossover (BCRC) for solving the problem which came out to be successful.

This work opts to choose both BCRC and SMCM crossovers. As they have proved to be better than others in studies as specified above, here an attempt is made evaluate them one against the other. The algorithm got coded in JAVA for both the instances with necessary changeover fields. The methodology is validated using the data obtained from a distribution firm.

2.0 Problem Background

The problem falls under the category of Capacitated Vehicle Routing Problem (CVRP). In CVRP there will be 'N' number of customers. Each customer will be geographically distributed at different locations with a demand 'd'. The demand at each location should be greater than zero. There will be a central depot from which the demand of all the customers will be served using homogeneous vehicle. The problem is to design a set of routes such that each route begins and ends at the same depot and each customer is serviced by exactly one vehicle. The sum of the demands of the customers assigned to a route should not exceed the vehicle capacity.

3.0 Mathematical Formulation

In finding the optimized routing solution the main aim is to minimize the total distance travelled and the total number of vehicles in meeting the customer requirements. The distance between the depot and customers and between customers one to one, shall be fetched from the distance matrix and the demand at each location shall be derived from the demand matrix. Therefore the objective function for the minimization of total distance travelled and for the minimization of number of vehicles shall be formed as Eq. (1) and Eq. (2) respectively given below;

$$\text{Min } D = \sum_{n=1}^N d_n \quad (1)$$

$$\text{where; } d_n = \sum_{i=0}^{N_n} d(i, i+1), n = 1, 2, \dots, N$$

$$\text{Min } N \quad (2)$$

- D = Total distance travelled by all vehicles
- d_n = Distance travelled by vehicle 'n'
- N = No of vehicles or routes
- N_n = number of customers visited by vehicle 'n'

4.0 Proposed Methodology

Genetic Algorithm (GA), being a population-based approach, is well apt to solve combined objective optimization problems. Genetic algorithms are based on ideas of evolution theory. The main principle here is that only the fittest entities survive. Genetic algorithms work with individuals, sometimes also called chromosomes, each representing a possible solution to a given problem [5]. In this work, genetic algorithm with fitness assignment procedure called Fitness Aggregated Genetic Algorithm is introduced for solving the CVRP.

The stages involved in the proposed methodology include; creation of initial population, calculate of fitness value, Selection, Crossover and Mutation.

The working of the proposed methodology is explained extracting a small portion of the original data collected for validation purpose. The distance and demand matrix is shown in Table 1 and Table 2 respectively.

Table 1. Demand Matrix

LOCATION	DEMAND
ETTUMANOOR	-
PALA	38
ERATTUPETTA	26
POONJR	10
TEEKOY	5
THODUPUZZHA	8
MOOLAMATTOM	22
KOTTAYAM	40
ERNAKULAM	32
VAIKOM	16

Table 2. Distance Matrix

	ETMR	PLA	ETPA	PNJR	TKY	TDPA	MLM	KTM	EKM	VKM
ETMR	0	16	27.9	32.8	32.8	45.6	53.2	10.9	55.2	32.5
PLA	16.4	0	12.3	17.2	17.2	30	37.6	27.4	66.7	43.4
ETPA	27.9	13	0	5.6	5.6	31.3	35.6	38.8	76.3	55.7
PNJR	32.8	18	5.6	0	6.4	35.4	39.7	43.8	84	60.6
TKY	32.8	17	5.6	6.4	0	31.8	36.2	43.8	84	60.7
TDPA	45.6	30	31.3	35.4	31.8	0	23.2	56.5	58.4	55.5
MLM	53.2	38	35.6	39.7	36.2	23.2	0	64.2	80	72.5
KTM	10.9	27	38.8	43.8	43.8	56.5	64.2	0	67.5	32.8
EKM	55.2	67	76.3	84	84	58.4	80	67.5	0	37.3
VKM	32.5	43	55.7	60.6	60.7	55.5	72.5	32.8	37.3	0

4.1 Stage 1: Creation of Initial Population

The initial population is created randomly as follows;

Firstly, any one customer is randomly selected and placed in the first location on the first route. Then another customer is selected randomly and if the capacity constraint is satisfied it is then assigned to the first route just after the previous customer. If the capacity constraint is not satisfied then the second route is initiated and this customer becomes the first customer on the second route. Now another different customer is randomly selected and if the capacity constraint is satisfied it is placed in the first route after the previous customer, else it is checked for capacity in the second route and if both these possibilities are out of question, then a new route is created and this customer is placed in the first location on the new route. The process is repeated till all the customers in the problem are assigned to routes. The pool of 10 initial solutions created as per methodology described is shown in Table 3.

Table 3. Initial population

Sol. No	R1	R2	R3	R4	R5	R6
1	KTM,TKY	EKM,TDP	MLM,VKM	PLA	PNJ	ETPA
2	PNJ,ETPA,TDP	KTM,TKY	VKM,MLM	EKM	PLA	
3	TKY,TDP,PNJ,MLM	KTY	ETPA,VKM	EKM	PLA	
4	ETPA,TKY,TDP	KTM	MLM,PNJ	VKM	EKM	PLA
5	TDP,VKM,MLM	EKM,PNJ,TKY	ETPA,	PLA	KTM	
6	MLM,VKM,TKY	PLA,TDP	EKM,PNJ	ETPA	KTM	
7	TDP,PLA	VKM,PNJ,TKY	EKM	KTM	MLM	ETPA
8	PLA,TDP	MLM,VKM,TKY	KTM	ETPA,PNJ	EKM	
9	KTM,TKY	TDP,MLM,PNJ	EKM	ETPA,VKM	PLA	
10	PLA,TKY	EKM,PNJ	ETPA,TDP	MLM,VKM	KTM	

Conditions applicable;

- Population size, S = 10; Max Vehicle Capacity = 47 units

4.2 Stage 2: Fitness Aggregation

As this work deals with combined objective optimization, fitness assignment cannot be done directly. Therefore GA with fitness assignment procedure known as Fitness Aggregated Genetic Algorithm (FAGA) is employed. The aim of this method is to find out the fitness value for multiple objectives. Firstly the total distance travelled and the number of vehicles is determined for each solution in the population. Then the fitness for travelled distance alone is calculated for all the solutions in a population by using the formula;

$$F(D)_i = \frac{(D)_{\max} - (D)_i}{(D)_{\max} - (D)_{\min}}; \quad i = 1, 2, \dots, S \quad (3)$$

Where; $F(D)_i$ is the fitness value of distance travelled for i^{th} solution in the population. $(D)_{\max}$ is the maximum distance travelled in the population. $(D)_{\min}$ is the the minimum distance travelled in the population. $(D)_i$ is the distance travelled for the i^{th} solution in the population. 'S' stands for the size of the population. $F(D)$ is calculated for all the solutions in the population.

Similarly the fitness for number of vehicles used is calculated for all the solutions using the formula;

$$F(N)_i = \frac{(N)_{\max} - (N)_i}{(N)_{\max} - (N)_{\min}}; \quad i = 1, 2, \dots, S \quad (4)$$

Where; $F(N)_i$ is the fitness value of number of vehicles used for i^{th} solution in the population. $(N)_{\max}$ is maximum number of vehicles used in the population. $(N)_{\min}$ is minimum number of vehicles used in the population. $(N)_i$ is the

number of vehicles used in the i^{th} solution in the population. $F(N)$ is calculated for all the solutions in the population.

Thereafter the aggregate fitness value (AFV) for a solution is determined by the formula;

$$F(D, N)_i = \frac{F(D)_i + F(N)_i}{n}; \quad i = 1, 2, \dots, S \quad (5)$$

Where $n=2$ (for bi-objective model)

$F(D, N)$ is calculated for all the solutions in the population. Since the problem is a minimization problem the solution with the highest aggregate fitness function value is the optimal solution for the problem. Table 4 shows the values after fitness aggregation procedure.

Table 4. Fitness Aggregation

Sol. No	Distance (D)	No of Routes (N)	F(D)	F(N)	F(D,N)
1	559	6	0.0212	0	0.0106
2	504	5	0.606	1	0.803
3	474	5	0.925	1	0.9625
4	467	6	1.0	0	0.5
5	516	5	0.478	1	0.739
6	561	5	0	1	0.5
7	519	6	0.446	0	0.223
8	510	5	0.542	1	0.771
9	488	5	0.776	1	0.888
10	523	5	0.404	1	0.702

4.3 Stage 3: Tournament Selection

Though different methods of selection are available, in this work ‘Tournament Selection method’ is chosen. Tournament selection is done so as to select from solutions of initial population and to introduce them on to the mating pool for ‘CROSS OVER’ operator to be applied in the next stage. The selection is based on the selection probability (P_s) and Average Fitness Value (AFV).

4.4 Stage 4: Crossover

Once the mating pool is generated, the next step is to implement cross over. Crossover in biological terms refers to the blending of chromosomes from the parents to produce new chromosomes for the offspring. The GA selects two strings at random from the mating pool. The GA then checks whether crossover should take place using a parameter called the crossover probability/ crossover rate. The crossover rate defines how often crossover should be applied [6]. In this work two types of crossovers, BCRC (Best Cost Rout Cross over) & SMCM (Sub Route Mapped Crossover Method) are considered.

BCRC Cross Over

In BCRC one route is selected each from both the parents randomly. Then the customers in the selected routs of parents are removed from one another. Then the removed customers from the parents are reinserted one by one to the respective parents at the best possible positions satisfying all the constraints. Fig. 1 shows an example of BCRC crossover.

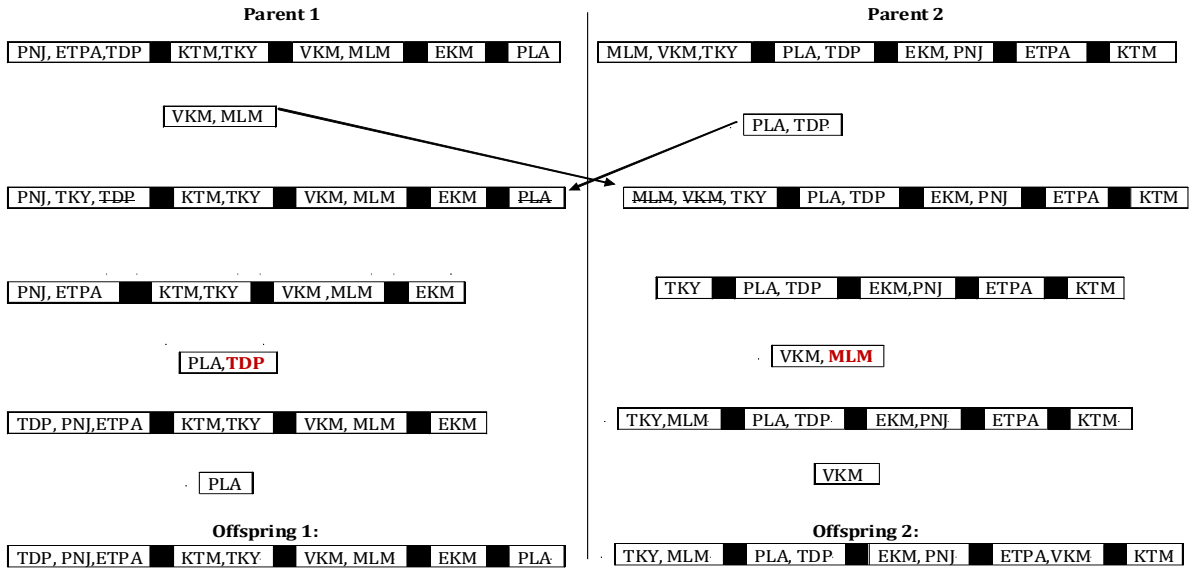


Fig.1 BCRC Crossover (Example)

SMCM Cross Over

In SMCM cross over, the last customers (nodes) of the different routes of the two parents becomes the first route of the two offspring's, subjective to capacity constraints. If the capacity constraints could not be met then a new route shall be initiated. Now remove the customers in the formed routes of one offspring from the parent of the other, for both. Then, copy down the remaining routes from both the parents. Now the last nodes of the remaining routes of both the parents are exchanged and inserted, satisfying capacity constraints and minimum total distance. Repeat the process till all the routes and its respective customers get exhausted. The process is shown in fig.2.

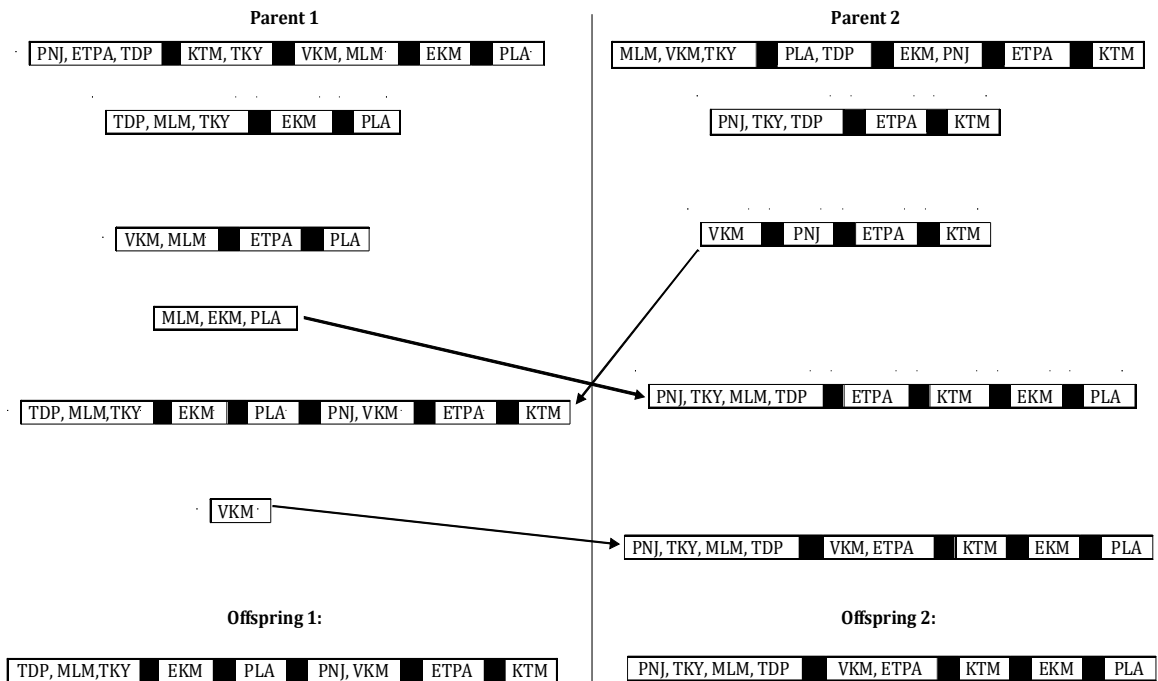


Fig.2 SMCM Crossover (Example)

4.4 Stage 4: Mutation

The offspring's formed in stage 4 is subjected to mutation. A mutation operator extracts a subset of nodes from a solution in the defined ways and reinserts them back by applying a random insertion [7]. It is applied with a mutation probability of ' P_m '. Mutation probability is generally set very low since it may cause duplication of good solutions. In this work Swap Mutation (SM) is chosen for implementation. Swap mutation encompasses the following steps;

An offspring generated after cross over is selected. One customer each from either of the two routes is randomly chosen. The location of the chosen customers is exchanged if it satisfies all the constraints. Example shown in fig 3.

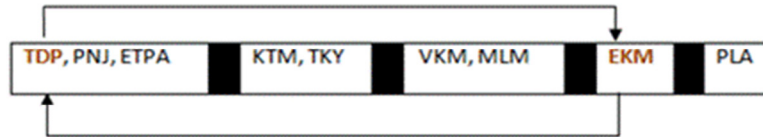


Fig.3 SWAP mutation

5.0 Analysis and Interpretation

Since earlier studies have proved that SMCM and BCRC are successful in generating feasible solutions for CVRP's compared to standard genetic operators like one point crossover, two point ordered crossover, uniform ordered crossover etc., the analysis and interpretation phase of this work starts with the comparison of these two proved crossovers.

5.1 Comparison of Crossovers

The algorithm got coded in JAVA for both the instances with necessary changeover fields and got executed on the full data. Initially the algorithm is executed using both the crossover operators separately for 100, 200, 300, 400 and 500 generations with the intention of exploring the better of the two. The execution was done on the following arbitrarily chosen parameters;

Selection probability; $P_s = 0.7$; Crossover probability; $P_c = 0.9$; Mutation Probability; $P_m = 0.1$

The result of the execution with both the crossovers is depicted graphically below.

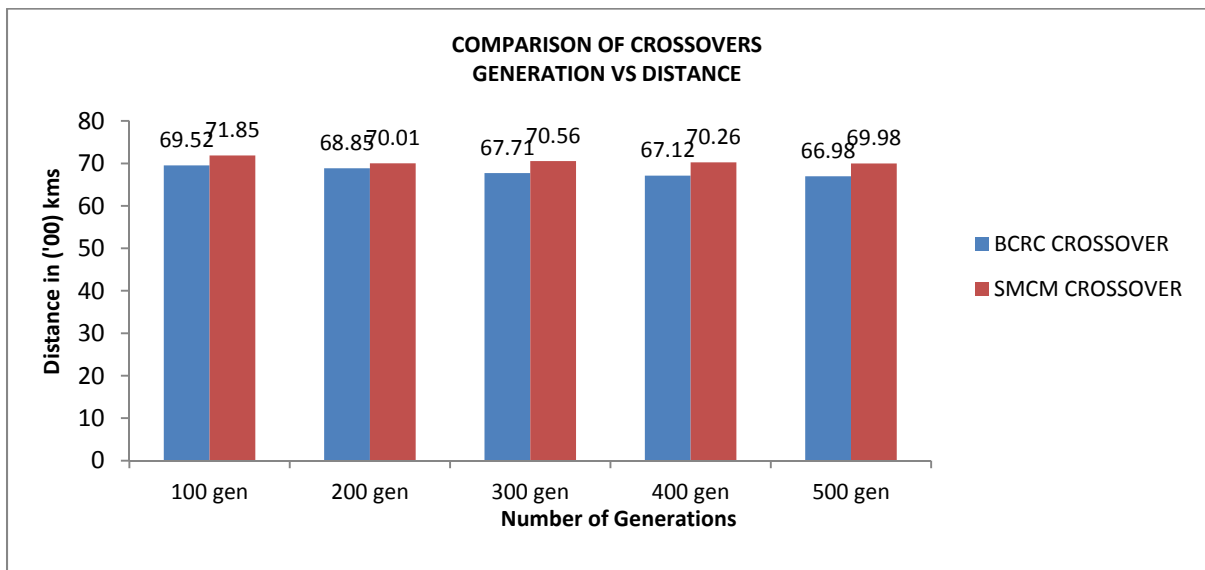


Fig.4 Comparison of crossovers

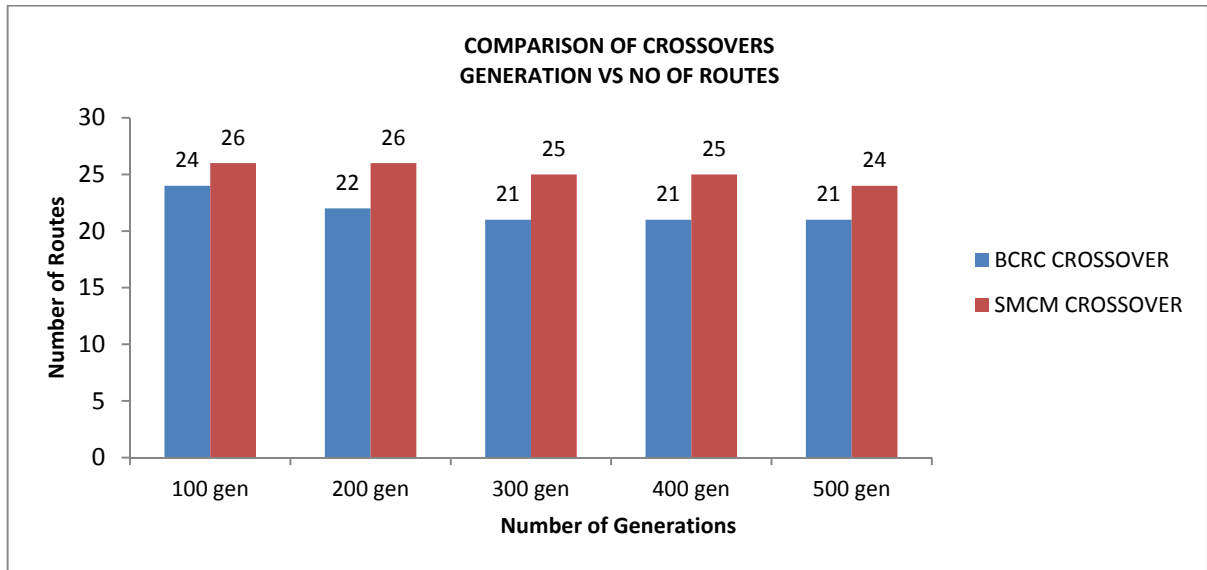


Fig.5 Comparison of crossovers

In each execution with BCRC crossover the results shows progressive improvements whereas in the case of SMCM the progression is not linear and the degree of optimization in successive generations is not exemplary, as evident from fig. 4 and fig.5 particularly when gauged with BCRC. So the outcome of the analysis favours the adoption of BCRC crossover.

5.2 Parameter Evaluation and Selection

In this study three different combinations are chosen randomly and were executed for 100 generations. The different randomly chosen combinations were;

Combination 1: $P_m = 0.01$; $P_s = 0.7$; $P_c = 0.8$;

Combination 2: $P_m = 0.01$; $P_s = 0.8$; $P_c = 0.9$

Combination 3: $P_m = 0.1$; $P_s = 0.9$; $P_c = 0.7$

The results obtained for each of the combinations is shown graphically below in fig. 6.

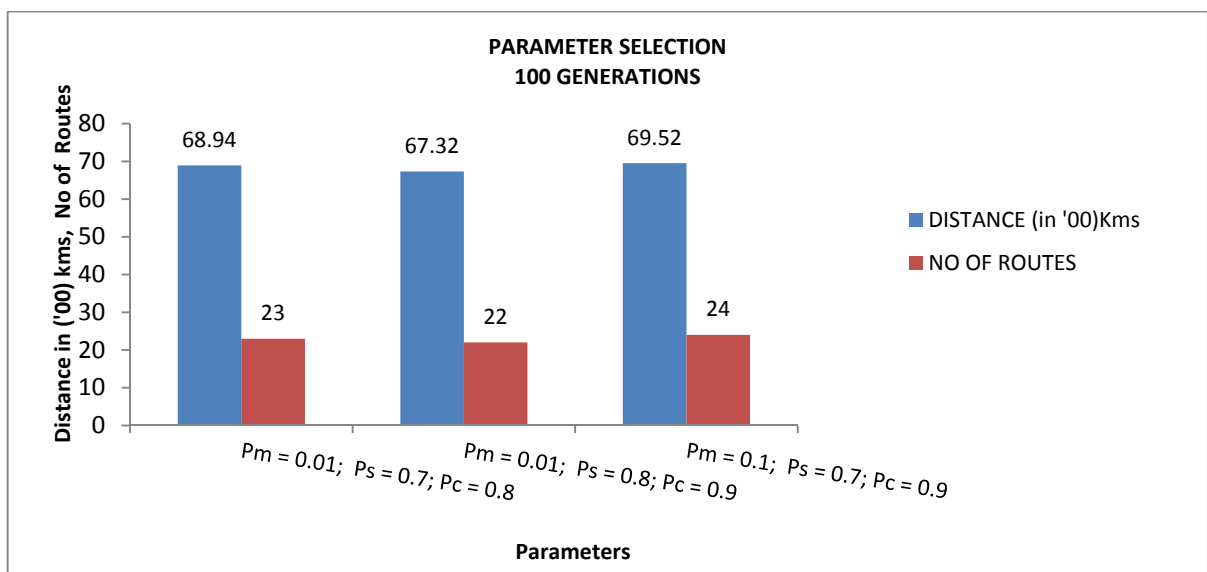


Fig.6 Parameter Selection

‘Combination 2’ with parameters $P_m = 0.01$; $P_s = 0.8$; $P_c = 0.9$ generated best results. ‘Combination 3’ proved to be the worst one. Since the type of crossover and the parameter combination to be coupled with it has become evident, the platform for execution is all set.

6.0 Result

The implementation of the algorithm was made by utilizing JAVA. The device specifications are 2.4 GHz Core i3 processor with 4GB RAM and Windows 10 operating system. The execution was made on the following parameters on the data received from the firm;

Maximum vehicle capacity = 60

Population size = 49

Number of generations = 5000

Crossover probability; $P_c = 0.9$

Selection probability; $P_s = 0.8$

Mutation Probability: $P_m = 0.01$

The variation in distance and number of vehicles observed in successive generations is given below in fig. 7;

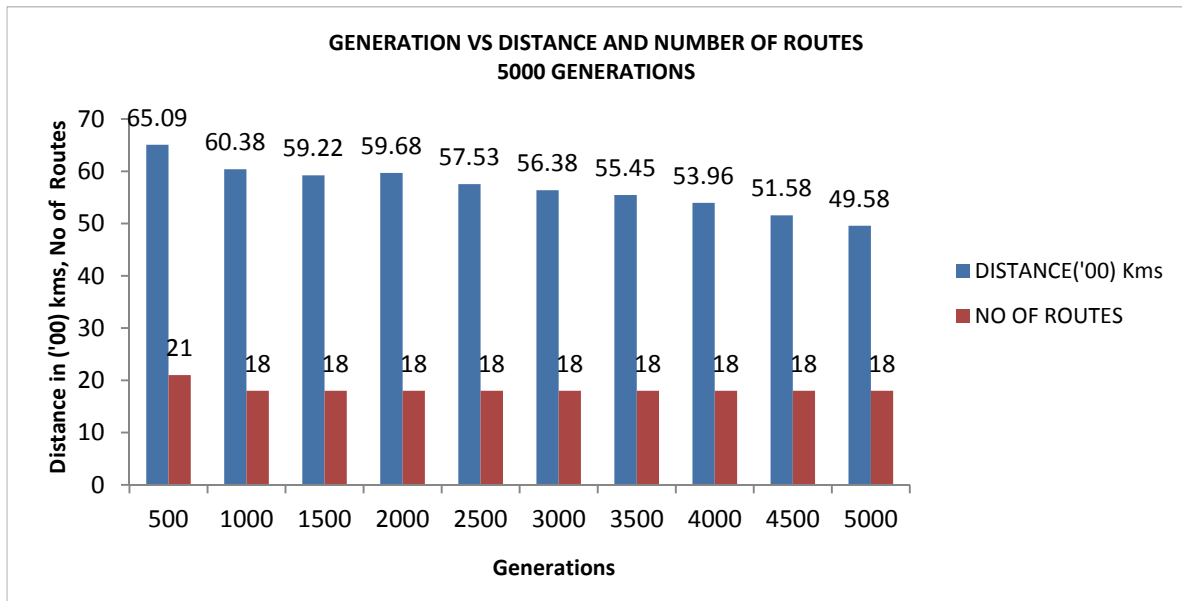


Fig. 7 Generation vs. distance & number of routes

The best solution obtained after 5000 generations is of 4958 kilometres and 18 routes. The individual configuration of these 18 routes of the best solution is shown in Table 5 and diagrammatically represented in fig 8.

Table 5. Best solution – Root sequences

ROUTES	SEQUENCE
R1	0 – 42 – 17 – 18 – 19 – 0
R2	0 – 26 – 27 – 29 – 0
R3	0 – 5 – 37 – 48 – 0
R4	0 – 45 – 46 – 0
R5	0 – 20 – 0
R6	0 – 23 – 43 – 31 – 30 – 0
R7	0 – 3 – 24 – 28 – 0
R8	0 – 38 – 22 – 0
R9	0 – 44 – 49 – 33 – 0
R10	0 – 40 – 16 – 32 – 36 – 35 – 0
R11	0 – 6 – 8 – 0
R12	0 – 39 – 34 – 0
R13	0 – 47 – 25 – 0
R14	0 – 1 – 4 – 13 – 7 – 0
R15	0 – 41 – 21 – 0
R16	0 – 2 – 9 – 0
R17	0 – 10 – 12 – 0
R18	0 – 11 – 15 – 14 – 0

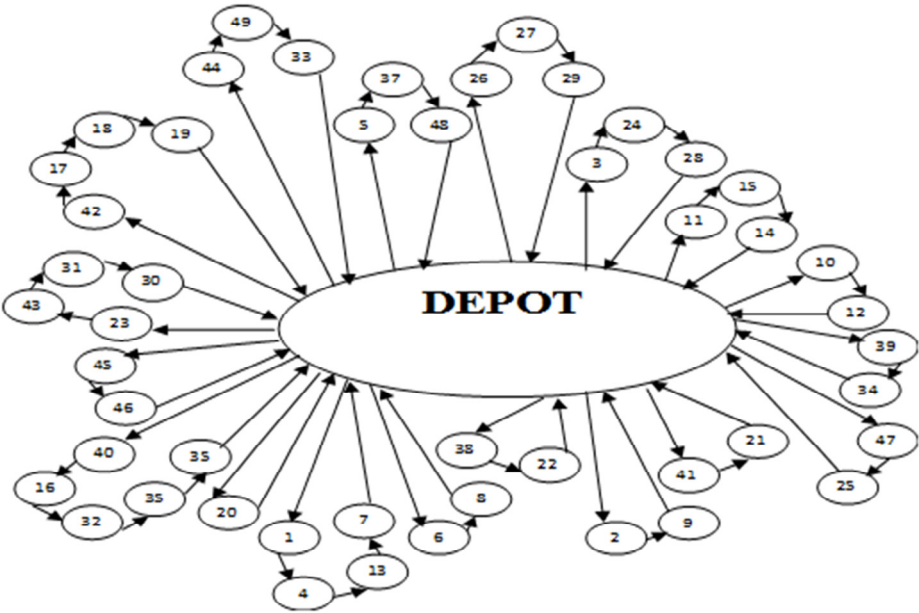


Fig. 8 Pictorial Representation of best solution

6.0 Methodology Validation

Since the data utilized for testing the efficacy of the model is obtained from a distribution firm, the benefit of implementing the model in the respective firm is worth analyzing. The consolidated cost data associated with operating each vehicle as obtained from the firm is shown in Table 6.

Table 6. Consolidated Cost data

Unit	Per vehicle	Per kilometer
Amount (Rs.)	900	5.91

The change in the number of vehicles and the total travelled distance observed after implementation is given in the figures 9 (a) & (b) below;

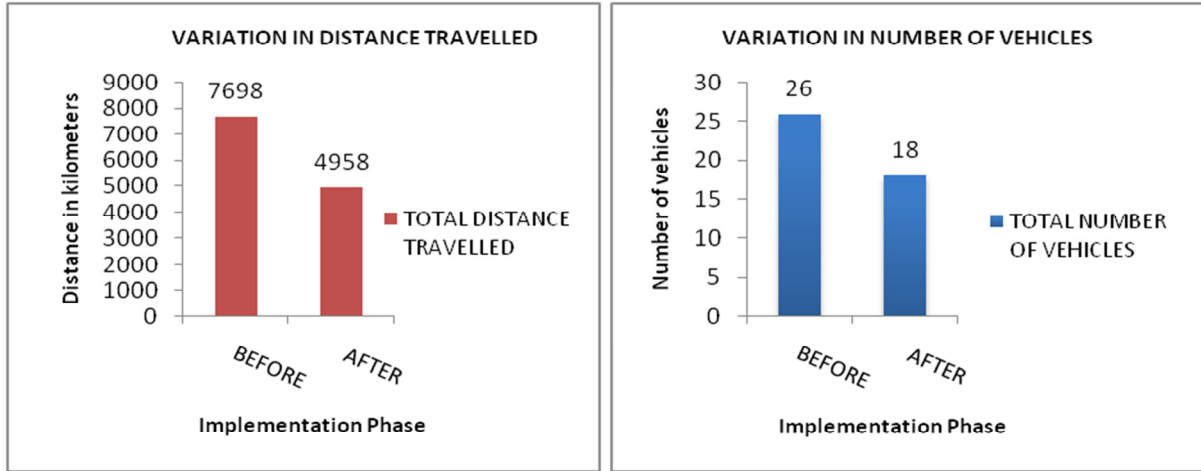


Fig. 9 (a) variation in distance; (b) variation in vehicles

Total savings to the firm in terms of rupees is shown in Table 7.

Table 7. Total savings

	Running cost (Rs.) (per month)	Initial expense (Rs.)
Before Implementation	1722379.5	16577314
After Implementation	1137544.5	11476602
Savings	584835.4	5100712

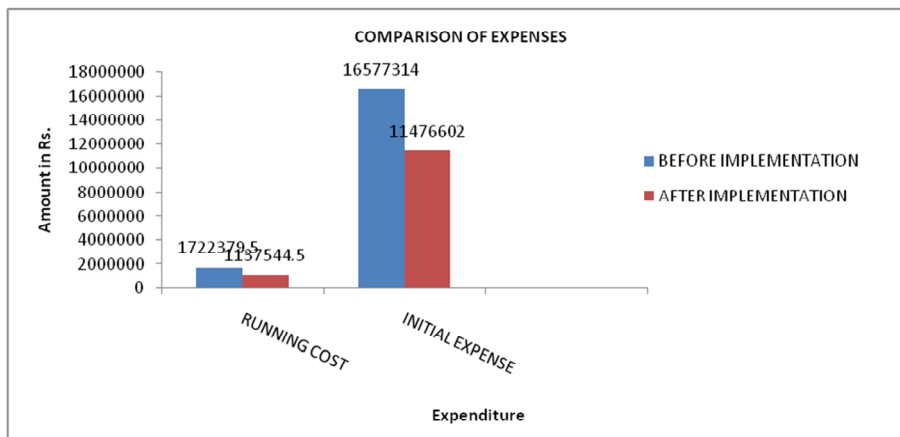


Fig. 10 Comparison of expenses

Therefore the proposed algorithm is of huge potential in exploring the optimized routing solution available for the firm and the numerical figures indicates that it shall benefit the firm in monetary terms to a great extend as evident from fig.10.

7.0 Conclusion

This work introduces a GA based approach for solving combined objective Capacitated Vehicle Routing Problems. The approach was tested on a real instance with population size 49 for 5000 generations. The proposed technique was able to converge to an optimal solution quickly irrespective of the population size and the number of generations opted. The work also compares BCRC and SMCM crossover's and proposes BCRC over SMCM, as the crossover operator. The problem was solved using both the crossover operators in the initial stage for 100 generations and the results proved that BCRC outperforms SMCM in terms of efficiency. Therefore BCRC crossover is chosen for solving the problem in this work. Parameter combinations to be implemented are found out by running three different sets of combinations, chosen arbitrarily. The combination of 0.01, 0.8 & 0.9 for mutation, selection and crossover probabilities respectively emerged out to be the best. The algorithm was coded utilizing JAVA and was run in Intel Core i3 2 MHz PC with 500 GB memory.

The proposed methodology holds a wide scope of applicability in vehicle routing issues. Further improvements are worthy of exploration in the future and might evolve useful.

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